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Isothermal oxidation behaviour of Nb–W–Cr Alloys

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ABSTRACT

A study of the effect of Cr content on the microstructure and isothermal oxidation behaviour of four alloys from the Nb–Cr–W system has been performed. Selection of specific alloy compositions has been based on the ternary isothermal sections. Oxidation experiments were conducted in air at 900 and 1300 °C for 24 h under isothermal conditions. Weight gain per unit area as function of the temperature has been used to evaluate the oxidation resistance. The phases present in the alloys and the oxide scales were characterized by XRD, SEM, and EDS. Microstructure consists of Nb solid solution and $NbCr_2$, Laves phase. The oxidation kinetics follows a parabolic behaviour at 1300 °C; the addition of 30% Cr resulted in the significant reduction of the parabolic oxidation rate. At 900 °C, alloys with higher Cr content exhibit higher oxidation rates in comparison to alloys with lower Cr content. The oxidation products are a mixture of $CrNbO_4$ and Nb_2O_5 and the amount of each oxide present in the mixture is related to the intermetallic phase content and the oxidation temperature. The characterization results delineate the effect of the Cr content on the oxidation mechanisms of these alloys that represent a promising base for high-temperature alloy development.

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1. Introduction

Materials that can withstand high temperatures and corrosive environments are constantly sought after in the power generation and aerospace industries. The use of refractory metals in high-temperature applications has been attractive for several years. Niobium-based alloys could meet the high-temperature capability envisaged to exceed the application temperatures of Ni base superalloys because of their low density, high melting points and high strength at high temperatures. However, the widespread use of niobium-based materials in aircraft, missiles, and certain reactors is limited because its oxidation resistance is still inadequate for structural applications.

Recent studies are oriented towards the development of niobium systems with Al, Cr, Ge, Hf, Si, and Ti additions; often referred to as refractory metal–intermetallic composites (RMICs) [1–12]. These niobium alloys contain multiphase microstructures comprised of silicides, $NbCr_2$, and Nb solid solution. A multiphase system is expected to provide a more favourable balance of high-temperature strength and good oxidation resistance. The system Nb–W–Cr is of interest because a duplex microstructure consisting of $NbCr_2$ and a bcc solid solution can be obtained. The low temperature fracture toughness can be improved by the Nb solid solution (Nb_{ss}) and the high-temperature strength and good oxidation resistance can be anticipated from the Laves phase [13,14]. The addition of Cr has been found to improve the oxidation resistance which has been attributed to the stabilization of the $NbCr_2$ [9]. The results of oxidation resistance in air for the alloys Nb–20W–5Cr and Nb–20W–10Cr have been reported by Kakarlapudi and Varma and Portillo et al. [15,16]. The study indicates that alloys containing higher chromium concentration and second phase intermetallic particles ($NbCr_2$), exhibit enhanced oxidation resistance than alloys without second phase particles. The oxidation of alloys from the Nb–W–Cr system is not well understood and it is affected by many factors including composition, microstructure and oxidation temperature.

Understanding the oxidation kinetics of Nb-based alloys is of crucial importance to improve the oxidation resistance. This article presents the results of an investigation performed in order to study the effect of the Cr content on the isothermal oxidation behaviour of four alloys from the Nb–W–Cr system at 900 and 1300 °C. The results are expected to provide direction towards the development of a structural material with improved oxidation resistance for use in high-temperature applications.

2. Experimental procedures

2.1. Alloys preparation

The alloys were fabricated by the Ames Laboratory of Iowa State University using arc melting technique. Nb, W, and Cr with at least 99.9% purity were melted on a water-cooled copper crucible in an atmosphere of high purity argon gas. The alloys were cast into

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Table 1
Nominal composition of Nb–Cr–W alloys.

Alloy identification	Composition (wt.%)			Composition (% atomic)		
	Nb	Cr	W	Nb	Cr	W
15Cr	75	15	10	70.2	25.1	4.7
20Cr	70	20	10	63.2	32.2	4.6
25Cr	65	25	10	56.6	38.9	4.4
30Cr	60	30	10	50.6	45.2	4.2

ingots with dimensions of 50 × 50 × 6 mm; final dimensions were obtained using electro-discharge machining (EDM).

The selection of specific alloy compositions has been based on the ternary isothermal sections of Nb–Cr–W diagrams at 1000 and 1500 °C [17]. Table 1 presents the nominal compositions of the alloys used for this study.

2.2. Characterization of the alloys

The alloys in the as-cast condition were prepared by standard sample preparation techniques for metallographic analysis. Cross-sections were mounted, ground, polished and etched using a solution with a proportion 2:1:2 ml of HF, HNO_3, and HCl, respectively. Microstructure of the alloys was characterized by scanning electron microscopy (SEM) and energy dispersive X-ray spectroscopy (EDS) using a Hitachi S-4800 scanning electron microscope equipped with energy dispersive spectrometer and EDAX 5.21 Genesis Software. Phases of the alloys were identified by X-ray diffraction (XRD) using a Bruker D8 Discover diffractometer with a voltage and a current of 40 kV and 40 mA, respectively and Cu Kα radiation (λ = 1.54 Å).

2.3. Oxidation experiments

The oxidation coupons with dimensions of approximately 4 × 4 × 3 mm were ground to 600 grit and ultrasonically cleaned in ethanol before exposure. Mass gain per unit area as a function of the temperature has been used to determine the alloy's oxidation resistance. Isothermal oxidation tests were carried out in static air at 900 and 1300 °C for 24 h using a Setaram LabSys Evo thermogravimetric analyzer with a heating rate of 23 °C/min. The weight gain vs. temperature graphs were obtained using Calisto data acquisition software v1.053.

The oxidation products were characterized by XRD with Cu Kα radiation. The surface morphologies of the oxide and the cross section of the scales were observed and analyzed by SEM and EDS.

3. Results and discussion

3.1. As-cast characterization

Backscattered electron images (BSE) of the alloys microstructure in the as-cast condition are presented in Fig. 1. The microstructure of alloys consists of Nb solid solution phase regions (white) and the intermetallic $NbCr_2$ (gray). It can be observed that the intermetallic phase increases with an increase in the Cr concentration.

The EDS analysis on the SEM indicates the locations of Cr and Nb rich regions. The BSE image and X-ray maps presented in Fig. 2 exhibits the two phase structure for 20Cr alloy in the as-cast condition. Table 2 presents the chemical composition determined by EDS for each alloy in the as-cast condition, as well as the fraction of the $NbCr_2$ phase which was measured using image analysis and the BSE contrast.

The XRD patterns obtained from the as-cast structures are shown in Fig. 3. The peak identification confirms the presence of two phases, Nb solid solution (α) and the cubic type Laves phase, $NbCr_2$ (C15).

3.2. Oxidation behaviour

The oxidation rates of the alloys were measured at 900 and 1300 °C in static air. Fig. 4 shows the weight gain per unit area as a function of the exposure time at 1300 °C for the four alloys. The oxi-

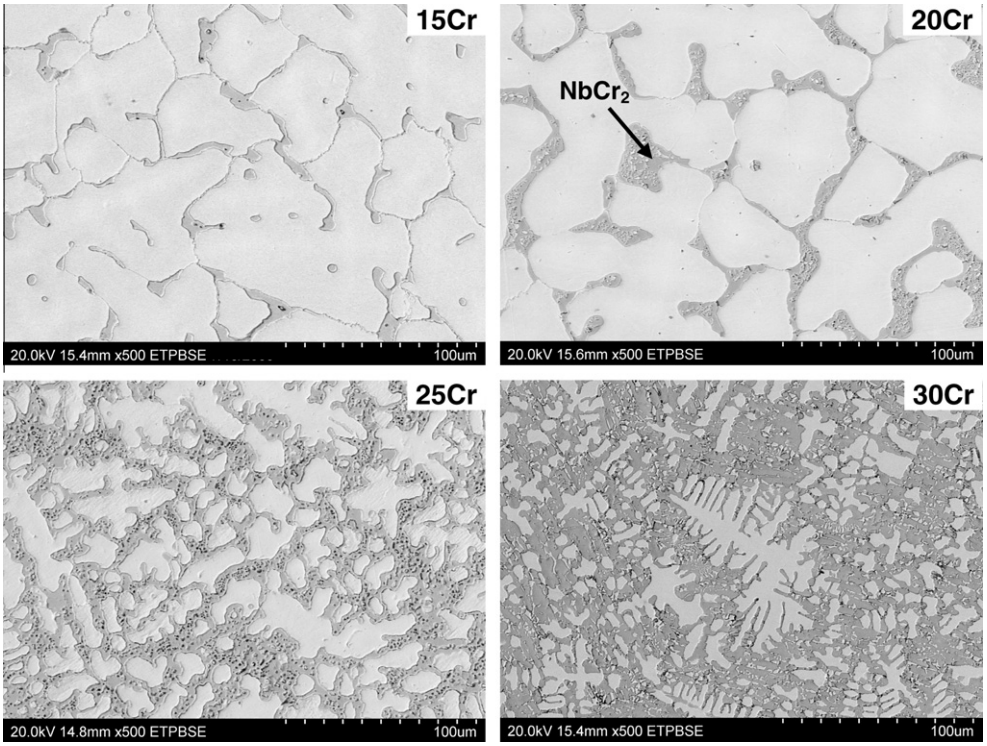


Fig. 1. BSE images of the as-cast microstructure of Nb–W–Cr alloys containing 15%, 20%, 25%, and 30% Cr.

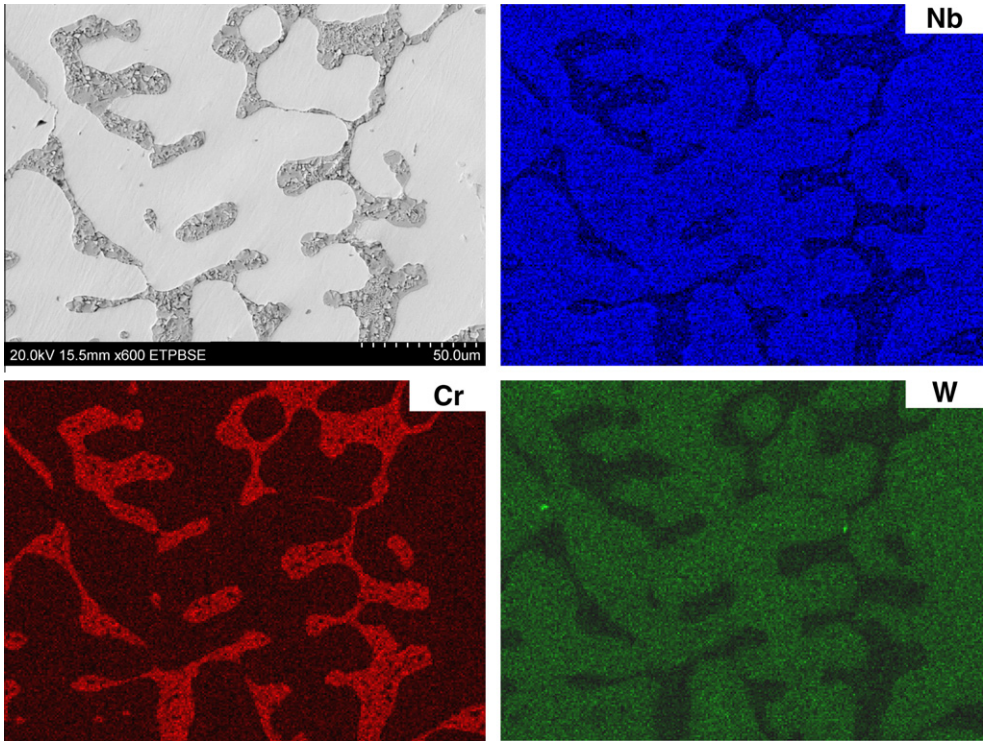


Fig. 2. BSE image and X-ray maps of 20Cr alloy showing Nb, Cr, and W distribution.

dation behaviour of all the alloys follows a parabolic law at this temperature; the time at which the oxidation rate stabilizes depends on the alloy composition. The alloys with higher volume percentage of $NbCr_2$ phase (25Cr and 30Cr) exhibited better oxidation resistance at 1300 °C. The alloy containing 15% Cr exhibits the lowest resistance with a weight gain that is about three times the weight gain of 30Cr alloy, which delineates the relationship between the oxidation resistance and the fraction of intermetallic phase in the alloy.

Table 2
Elemental compositions determined by EDS analysis and area fraction of the $NbCr_2$ phase in the as-cast condition.

Alloy	Element (wt.%)			$NbCr_2$ phase (area percentage)
	Nb	Cr	W	
15Cr	74.7–76.7	12.3–13.9	10.3–12.6	7.6 ± 0.9
20Cr	68.7–70.6	17.3–18.4	11.8–12.7	18.3 ± 1.2
25Cr	62.2–64.8	24.2–26.2	10.0–11.2	43.0 ± 0.8
30Cr	61.5–62.7	28.1–30.2	8.0–9.8	56.7 ± 0.9

The samples with 25% and 30% Cr showed poor oxidation resistance at 900 °C as shown in Fig. 5. The weight gain per unit area of the 30Cr alloy is approximately two times higher than the weight gained at 1300 °C in spite of a temperature difference of 400 °C. 15Cr and 20Cr alloys exhibit near identical oxidation behaviour at the initial oxidation stage (up to 12 h), and then the alloy containing 20% Cr exhibits a slightly better oxidation resistance which may be attributed to the higher Cr content.

The parabolic rate constants for oxidation tests are listed in Table 3. The parabolic oxidation rate constants (k_p) were calculated by least-square, linear-regression analysis of the mass gain per unit surface ($\Delta m/A$) and the exposure time (t),

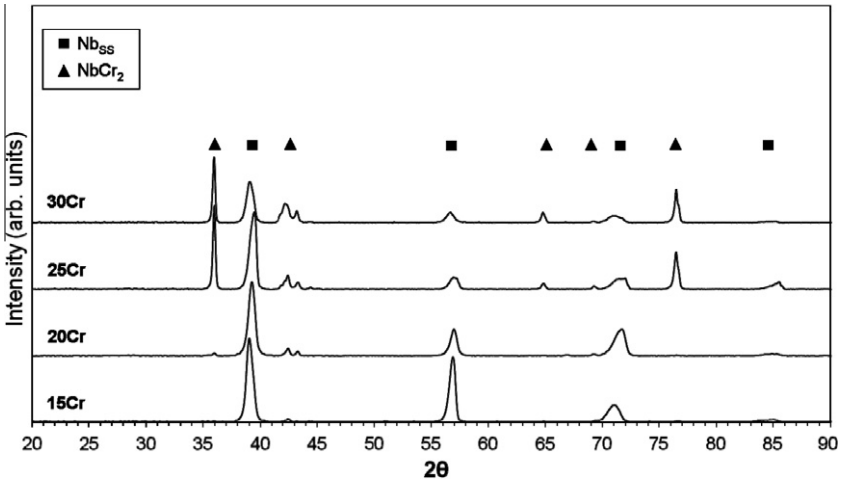


Fig. 3. XRD patterns of 15Cr, 20Cr, 25Cr, and 30Cr alloys in the as-cast condition.

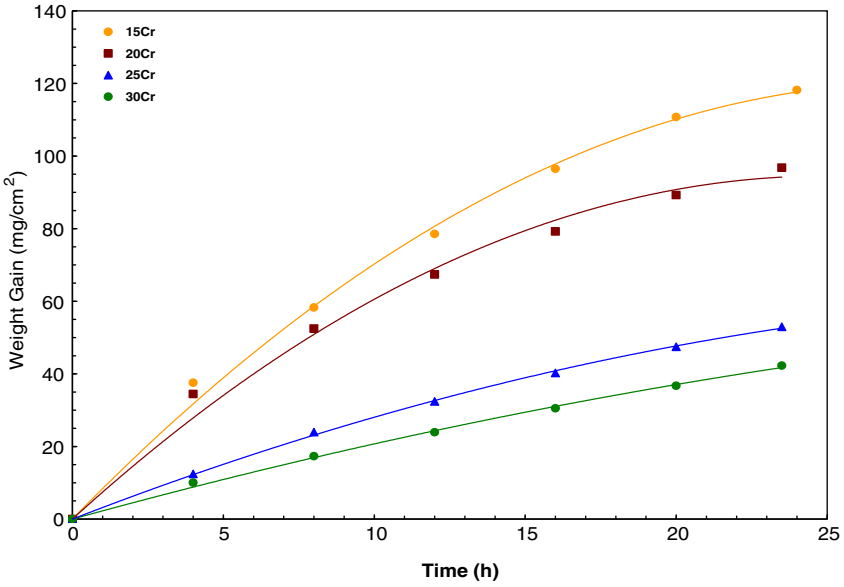


Fig. 4. Isothermal oxidation curves for 15Cr, 20Cr, 25Cr, and 30Cr alloys at 1300 °C.

$$\left(\frac{\Delta m}{A}\right)^2 = k_p t \quad (1)$$

At 900 °C the 25Cr and 30Cr alloys exhibit higher oxidation rates than the alloys containing less Cr, delineating the poor oxidation resistance of the alloys at this temperature. At 1300 °C the oxidation rate values increase with temperature; and decrease as the Cr concentration increases. The parabolic oxidation rate constant is reduced by 88% with the addition of 30% Cr instead of 15% Cr at 1300 °C.

Table 3
Values of parabolic rate constants (k_p) corresponding to isothermal exposure.

Temperature	Alloy	k_p (mg^2 cm^{-4} h^{-1})
900 °C	15Cr	96.1
	20Cr	82.2
	25Cr	333.5
	30Cr	308.9
1300 °C	15Cr	574.7
	20Cr	391.9
	25Cr	108.1
	30Cr	65.3

3.3. Oxide scale characterization

The oxidation products of the different alloys were identified by XRD. Figs. 6 and 7 present the XRD patterns obtained from the oxidation products of the alloys after 24 h of exposure at 900 and 1300 °C, respectively. The products are mainly a mixture of $CrNbO_4$, and Nb_2O_5. Two modifications of Nb_2O_5 were identified, one corresponding to the M-form (monoclinic) present at 900 °C, and the

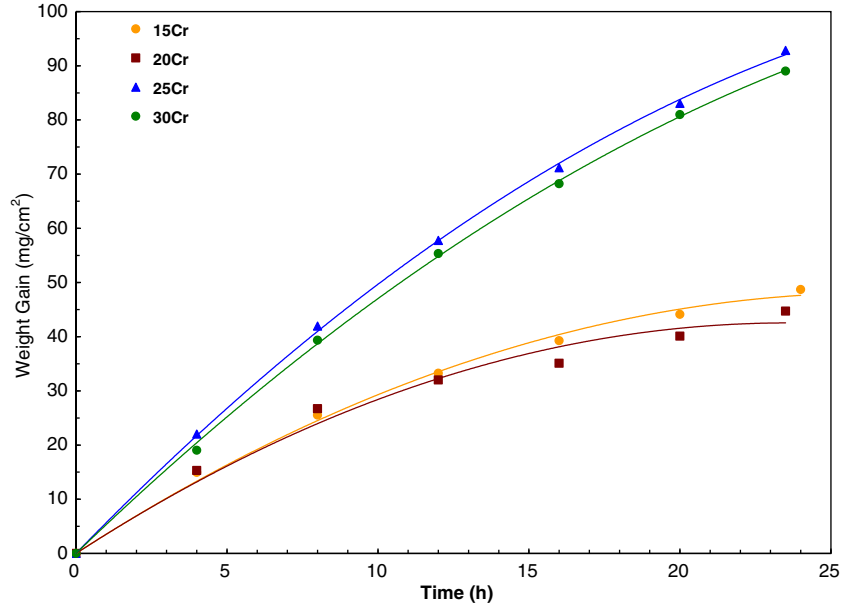


Fig. 5. Isothermal oxidation curves for 15Cr, 20Cr, 25Cr, and 30Cr alloys at 900 °C.

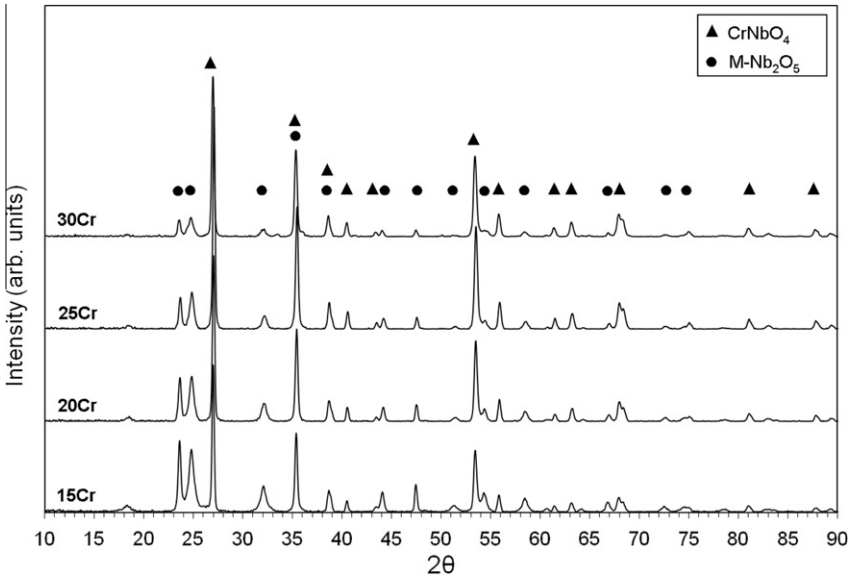


Fig. 6. XRD pattern of the oxidation products obtained from the four alloys after 24 h of exposure at 900 °C.

high-temperature H-form (monoclinic) observed at 1300 °C. While the types of oxides formed are similar, the relative amounts differ according to Cr content and temperature. As the Cr concentration increases the intensity of the Nb_2O_5 peaks decreases, thus $CrNbO_4$ is the predominant oxide formed in 30Cr sample as seen by the relative peak heights.

The intensity ratio of the most intense reflection for $CrNbO_4$ and Nb_2O_5 ($I_{CrNbO_4}/I_{Nb_2O_5}$) was used to perform a comparison of the relative amounts of $CrNbO_4$ and Nb_2O_5 in the oxidation product. The results are plotted as a function of the intermetallic phase volume percentage in Fig. 8. The relative intensity ratio increases with the volume percent of intermetallic phase, therefore it is reasonable to believe that the formation of $CrNbO_4$ is favoured in alloys with higher volume percent of Laves phase. The graph in Fig. 8 also shows the effect of temperature, the relative intensity ratio increases significantly in the case of alloys with higher intermetallic phase content, while the range of variation is small in the case of alloys with lower values of intermetallic phase volume percentage (15Cr and 20Cr).

Fig. 9a–d shows BSE images of the surface morphologies of the 15Cr and 30Cr alloys after oxidation at 900 and 1300 °C for 24 h. The characteristics of the scale formed on the surface change with Cr content and temperature. The samples with the best oxidation resistance at 900 °C (15Cr and 20Cr) exhibit smoother surfaces. The surface on 25Cr and 30Cr alloys is rough and prone to spallation, which is in agreement with the poor oxidation resistance observed. Fine granular oxide structures are observed at 1300 °C.

The oxide–metal interface images of sample 15Cr oxidized at 900 °C for 24 h are presented in Fig. 10. The oxidation products were identified based on the results from the XRD analysis, the EDS analysis and BSE contrast. The oxide scale is a combination of Nb_2O_5 (light gray areas) and $CrNbO_4$ (dark gray areas) in the form of alternated oxide layers. The oxide layers contain fissures parallel to the metal surface which are more evident at the $CrNbO_4$ rich areas. These types of irregularities are usually caused by the stress generated by the difference in the thermal expansion coefficients of the two oxides. The first oxide layer has a lighter appearance than the successive layers (Fig. 10a), EDS elemental analysis

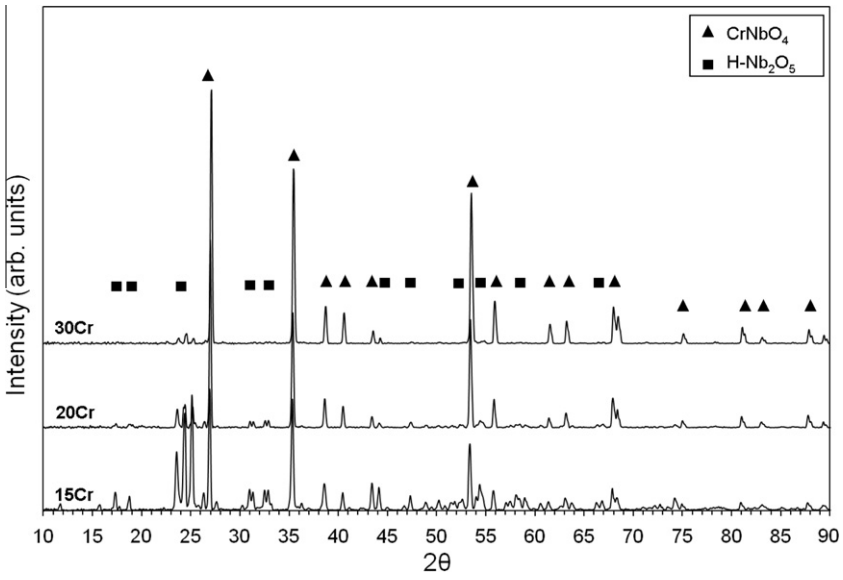


Fig. 7. XRD pattern of the oxidation products obtained from 15Cr, 20Cr, and 30Cr alloys after 24 h of exposure at 1300 °C.

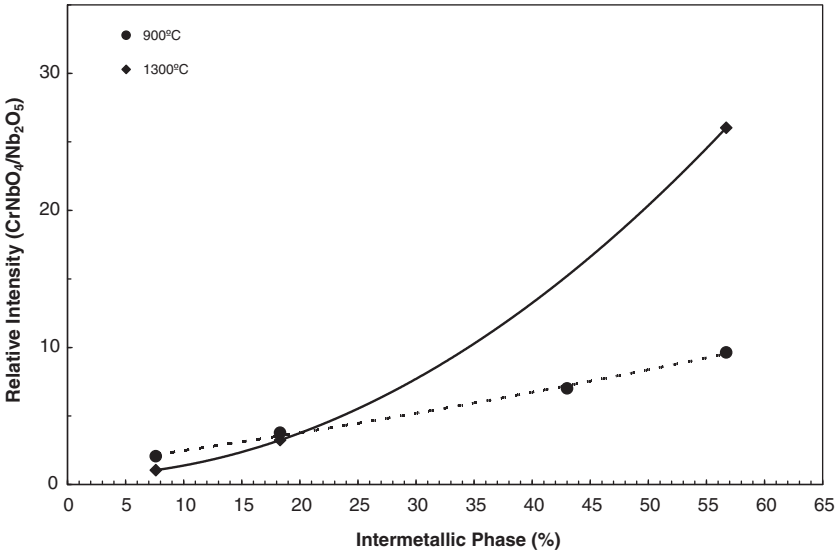


Fig. 8. Relative intensity ratio as a function of the volume percentage of intermetallic phase.

revealed a Nb/O ratio close to 1, which leads to believe that this layer is the intermediate reaction product NbO. The presence of the NbO at the metal–oxide interface depends on time, temperature, oxygen partial pressure, and the local radius of curvature of the metal–oxide interface [18,19]. The oxidation of Nb–Cr alloys resulting in alternate layers of Nb_2O_5 and $CrNbO_4$ has been previously reported [20]; these studies indicate that oxide growing occurs by inward oxygen diffusion.

The oxide–metal interfaces of 25Cr and 30Cr alloys oxidized at 900 °C presented numerous microcracks around the perimeter of the sample which caused alloy fragmentation before the complete oxidation of the phases. These microcracks cause the accelerated oxidation and powder formation observed for these alloys. The poor oxidation resistance of Nb-based alloys at intermediate temperatures caused by internal oxidation and microcrack formation has been previously reported by Bewlay et al. [2], Chan [7], Geng et al. [9,21], and Moricca and Varma [22].

The oxide–metal interface images of 15Cr alloy oxidized at 1300 °C for 24 h is presented in Fig. 11. The oxide scale on 15Cr alloy is a mixture of Nb_2O_5 (light gray areas) and $CrNbO_4$ (dark gray areas), however, there are no alternating layers or fissures as the ones observed at 900 °C; instead a more compact scale with small pores is formed. The distribution of $CrNbO_4$ appears to follow the distribution of the intermetallic phase indicating that the forma-

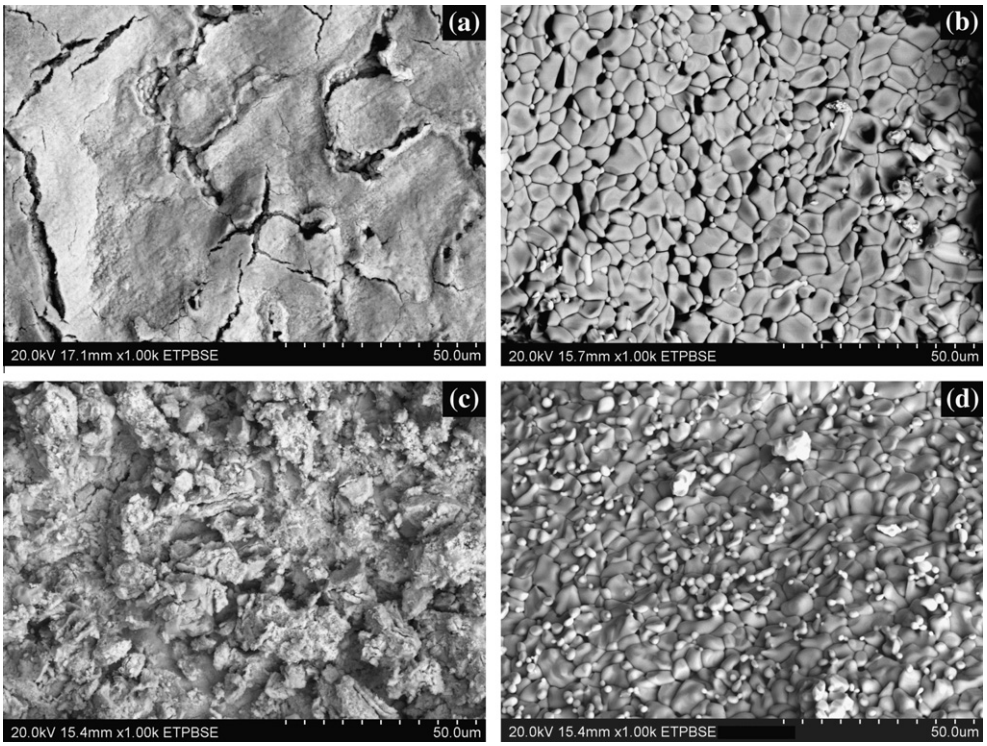


Fig. 9. BSE images of the surfaces of alloys oxidized for 24 h. (a) 15Cr at 900 °C, (b) 15Cr at 1300 °C, (c) 30Cr at 900 °C, and (d) 30Cr at 1300 °C.

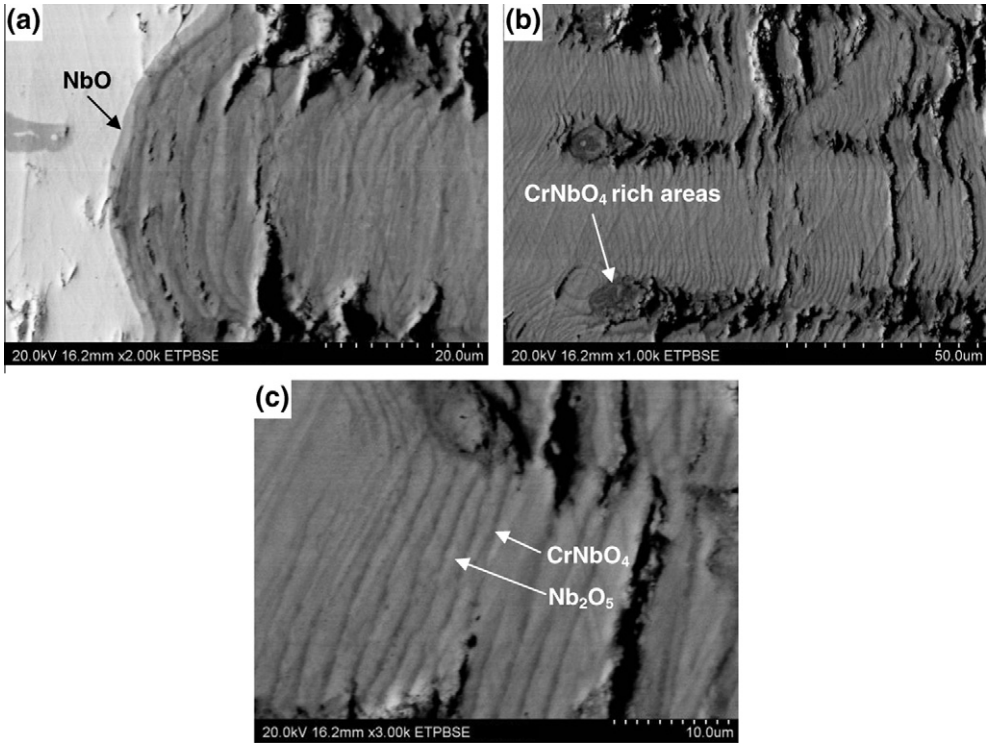


Fig. 10. BSE images of the metal–oxide interface of 15Cr alloy after 24 h of exposure at 900 °C. (a) Interface, (b) and (c) oxide scale.

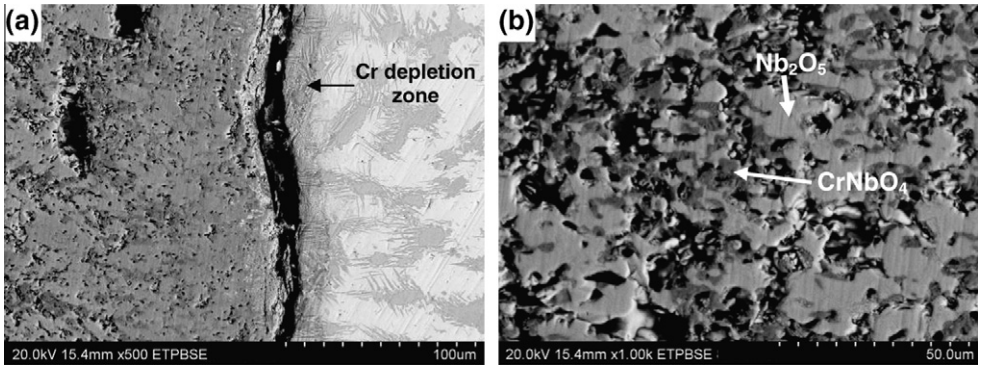


Fig. 11. Cross-section BSE images of the metal–oxide interface (a) and oxide layer (b) of 15Cr alloy after 24 h of exposure at 1300 °C.

tion of this oxide it is more frequent at the $NbCr_2$ regions. A Cr depletion layer of approximately 20 microns can be observed at the metal–oxide interface, which suggests that at this temperature the oxidation process occurs not only by inward diffusion of oxygen but also by means of outward diffusion of Cr. A Cr depletion zone is also present in 20Cr alloy and the oxide scale is a mixture of Nb_2O_5, $CrNbO_4$, and uniformly distributed pores.

The oxide scale of 25Cr and 30Cr alloys consists of multiple layers of approximately 100 microns with a mixture of Nb_2O_5 and $CrNbO_4$ oxides (Fig. 12) instead of a compact scale as observed for 15Cr and 20Cr alloys. Fig. 13 presents the metal–oxide interface and the oxide layers of 25Cr alloy oxidized at 1300 °C. The oxide scale spalls once it reaches a critical thickness and the separation of the oxide layer leaves a new metal surface for the formation of a new oxide layer. The metal–oxide interface of 25Cr and 30Cr alloys also exhibits a Cr depleted zone of approximately 50 microns indicating that oxidation occurs not only by oxygen inward diffusion but also by outward diffusion of Cr. A thin Cr rich oxide layer at the metal–oxide interface of around 10 microns is also observed. The EDS elemental analysis indicates that the Cr/O ratio is approximately 0.7, which leads to believe that this layer might be Cr_2O_3.

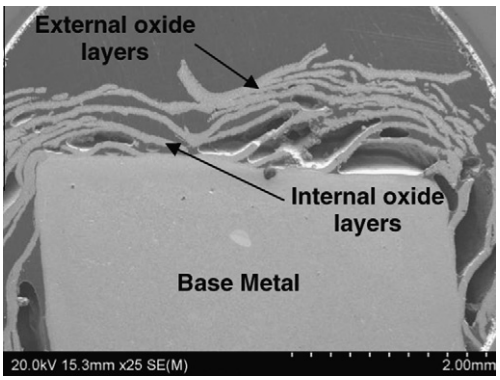


Fig. 12. SEM images of the oxide layers formed on 25Cr and 30Cr alloys after 24 h of exposure at 1300 °C.

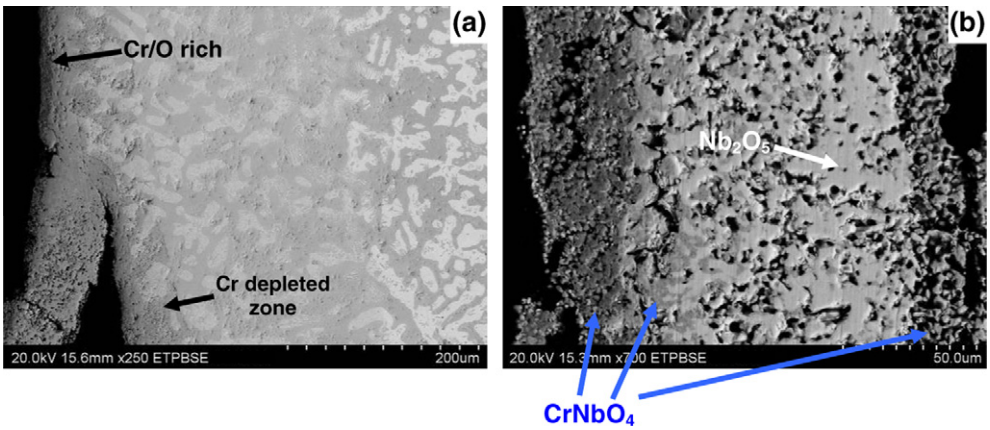


Fig. 13. Cross-section BSE images of the metal–oxide interface (a) and an external oxide layer (b) for 25Cr alloy after 24 h of exposure at 1300 °C.

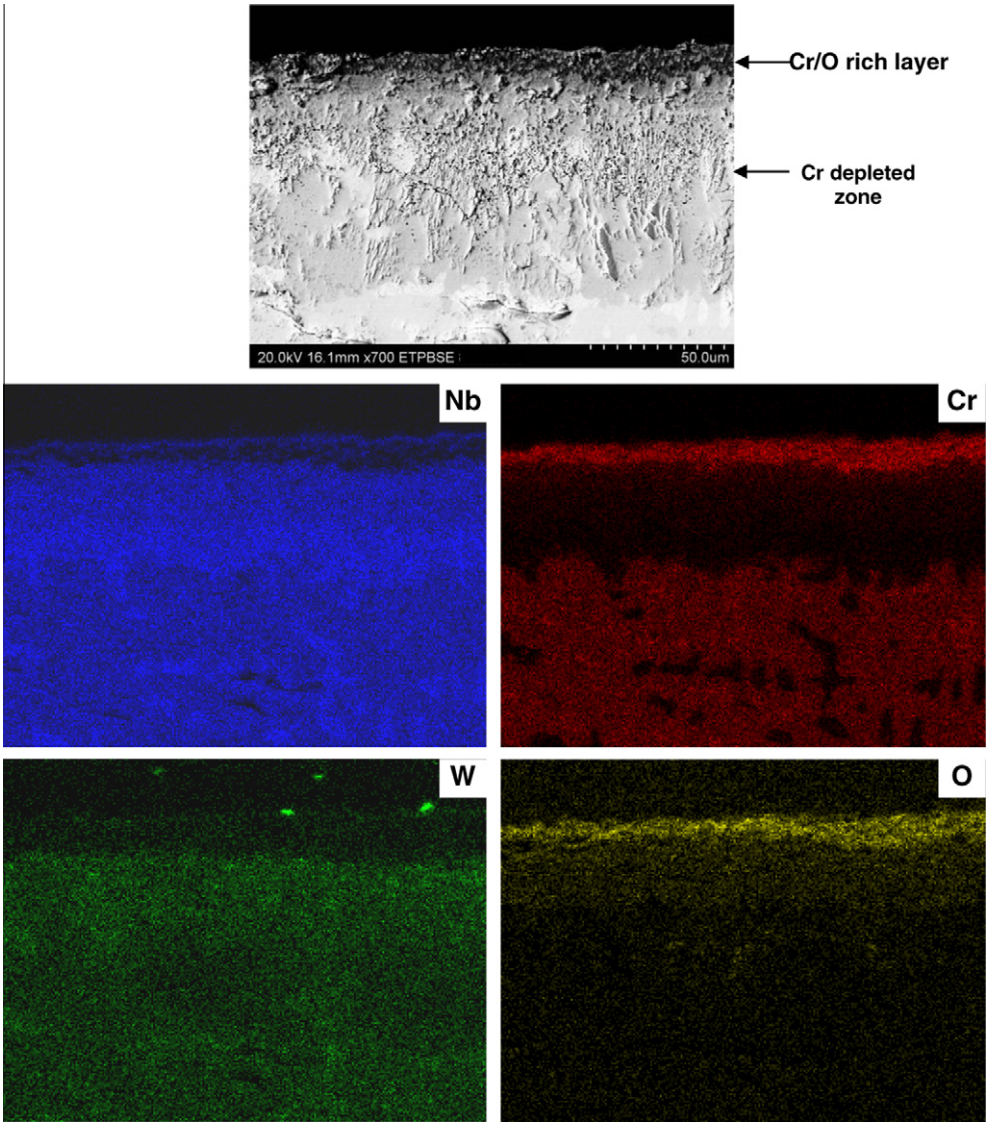


Fig. 14. BSE image and X-ray maps showing Nb, Cr, W, and O distribution for 30Cr alloy oxidized 24 h at 1300 °C.

However this assumption is only based on the EDS analysis, the XRD analysis did not reveal the presence of this oxide, but according to the thickness of this layer, the amount of this oxide might be too small to be detected in the oxide scales or stayed at the metal surface when the oxides spalled off during the analysis. Chromia (Cr_2O_3) evaporates in the form of CrO_3 at high temperatures, the

evaporation becomes important at temperatures higher than 1000 °C and near atmospheric oxygen partial pressures. The vaporization losses are reduced in static air since a large fraction of the evaporating molecules are reflected back on the surface, also the presence of other oxides on top of the Cr_2O_3 layers may inhibit the evaporative loss [19,23]. Evidence of the presence of Cr_2O_3 at the external oxide layers is not observed. The oxide layers are a mixture of $CrNbO_4$ and Nb_2O_5 as shown in Fig. 13. Fig. 14 presents a BSE image and the X-ray maps showing Nb, Cr, W, and O distribution for 30Cr alloy at the metal interface, Cr depletion area and Cr/O rich layer can be clearly observed.

4. Conclusions

Alloy microstructure consists of Nb solid solution phase surrounded by a network of the $NbCr_2$ Laves phase. The amount of intermetallic phase is related to the amount of Cr content.

The oxidation kinetics for the four alloys follow a parabolic behaviour at all the selected temperatures. Alloys containing lower Cr concentrations exhibit better oxidation resistance than alloys with higher Cr content at 900 °C; the unsatisfactory oxidation behaviour of 25Cr and 30Cr alloys is related to microcrack formation at the alloy perimeter. At 1300 °C, the Nb solid solution and the Laves phase oxidize cooperatively to form a scale with a mixture of $CrNbO_4$ and Nb_2O_5 oxides. The outward diffusion of Cr results in the formation of a depleted zone at the oxide–metal interface. The addition of 30% Cr resulted in the significant reduction of the parabolic oxidation rate at 1300 °C.

The oxidation products of the alloys are a mixture of $CrNbO_4$ and Nb_2O_5. Two modifications of Nb_2O_5 were identified, M-form at 900 °C and H-form at 1300 °C. Formation of $CrNbO_4$ is favoured in alloys with higher volume percent of Laves phase at high temperatures (1300 °C).

The oxidation behaviour of the alloys is controlled by the volume fraction of intermetallic phase. The results obtained delineate the complexity of the oxidation mechanism and the influence of microstructure and composition of the alloys.

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